

5.8.7

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Question

Yash scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each incorrect answer, then Yash would have scored 50 marks. How many questions were there in the test?

Solution

Let the number of right and wrong answers be x and y respectively.

$$\begin{pmatrix} 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 40 \quad (1)$$

$$\begin{pmatrix} 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 50 \quad (2)$$

These can be combined to give the matrix equation

$$\begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 40 \\ 50 \end{pmatrix} \quad (3)$$

Solution

This gives the augmented matrix

$$\left(\begin{array}{cc|c} 3 & -1 & 40 \\ 4 & -2 & 50 \end{array} \right) \xleftrightarrow{R_2 \leftarrow R_2 - \frac{4}{3}R_1} \left(\begin{array}{cc|c} 3 & -1 & 40 \\ 0 & -\frac{2}{3} & -\frac{10}{3} \end{array} \right) \quad (4)$$

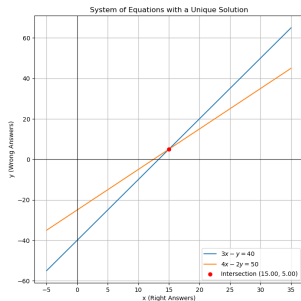
This gives the following values of x and y :

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 15 \\ 5 \end{pmatrix} \quad (5)$$

Thus, the number of questions in the test is

$$x + y = 20 \quad (6)$$

Plot



```
#include <stdio.h>

int solve_2x2(double A[4], double b[2], double x[2]) {
    double det = A[0]*A[3] - A[1]*A[2];
    if (det == 0.0) {
        return -1;
    }
    x[0] = (b[0]*A[3] - b[1]*A[1]) / det;
    x[1] = (A[0]*b[1] - A[2]*b[0]) / det;
    return 0;
}
```

Python + C Code

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes
lib = ctypes.CDLL("./code.so")
lib.solve_2x2.argtypes = [ctypes.POINTER(ctypes.c_double),
                           ctypes.POINTER(ctypes.c_double),
                           ctypes.POINTER(ctypes.c_double)]
lib.solve_2x2.restype = ctypes.c_int

A = np.array([[3, -1],
              [4, -2]], dtype=np.float64)
b = np.array([40, 50], dtype=np.float64)
x = np.zeros(2, dtype=np.float64)
status = lib.solve_2x2(A.ctypes.data_as(ctypes.POINTER(ctypes.
    c_double)),
                      b.ctypes.data_as(ctypes.POINTER(ctypes.
    c_double)),
                      x.ctypes.data_as(ctypes.POINTER(ctypes.
    c_double)))
```

```
if status == 0:
    x_sol, y_sol = x
    print(f"The system has a unique solution:")
    print(f"x = {x_sol}")
    print(f"y = {y_sol}")
    x_vals = np.linspace(x_sol - 40, x_sol + 40, 400)
    y1 = (3*x_vals - 40)
    y2 = (2*x_vals - 25)
    plt.figure(figsize=(8, 8))
    plt.plot(x_vals, y1, label=r'$3x - y = 40$')
    plt.plot(x_vals, y2, label=r'$4x - 2y = 50$')
    plt.plot(x_sol, y_sol, 'ro', label=f'Intersection ({x_sol:.2f}
        }, {y_sol:.2f})')
    plt.axhline(0, color='black', linewidth=0.8)
    plt.axvline(0, color='black', linewidth=0.8)
    plt.xlabel("x")
    plt.ylabel("y")
```

```
plt.title("System of Equations with a Unique Solution")
plt.legend()
plt.grid(True)
plt.savefig("../figs/Figure_1.png")
plt.show()
else:
    print("The system does not have a unique solution.")
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[3, -1],
              [4, -2]])
b = np.array([40, 50])
try:
    solution = np.linalg.solve(A, b)
    x_sol, y_sol = solution
    print(f"The system has a unique solution:")
    print(f"x = {x_sol}")
    print(f"y = {y_sol}")
    x_vals = np.linspace(x_sol - 40, x_sol + 40, 400)
    y1 = (3*x_vals - 40)
    y2 = (2*x_vals - 25)
    plt.figure(figsize=(8, 8))
    plt.plot(x_vals, y1, label=r'$3x - y = 40$')
    plt.plot(x_vals, y2, label=r'$4x - 2y = 50$')
```

```
plt.plot(x_sol, y_sol, 'ro', label=f'Intersection ({x_sol}, {  
    y_sol})')  
plt.axhline(0, color='black', linewidth=0.8)  
plt.axvline(0, color='black', linewidth=0.8)  
plt.xlabel("x")  
plt.ylabel("y")  
plt.title("System of Equations with a Unique Solution")  
plt.legend()  
plt.grid(True)  
plt.savefig("../figs/Figure_2.png")  
plt.show()  
  
except np.linalg.LinAlgError:  
    print("The system does not have a unique solution.")
```